pyDMS

Reproducible Dual–Mode Sorption Parameters

Analysis performed at 2025-12-16 17:02:54 Eastern Standard Time

DMS Parameters

Temperature	Parameter	Result ± Std. Dev.
308 K	k_D	3.38947 ± 0.01562
	C'_H	67.26171 ± 0.22637
	b	0.64972 ± 0.00323
318 K	k_D	2.76039 ± 0.00718
	C'_H	63.79970 ± 0.12341
	b	0.48237 ± 0.00225
328 K	k_D	2.27639 ± 0.01225
	C'_H	61.41546 ± 0.28103
	b	0.36468 ± 0.00187
338 K	k_D	1.89879 ± 0.01714
	C'_H	60.06897 ± 0.46989
	b	0.28031 ± 0.00169

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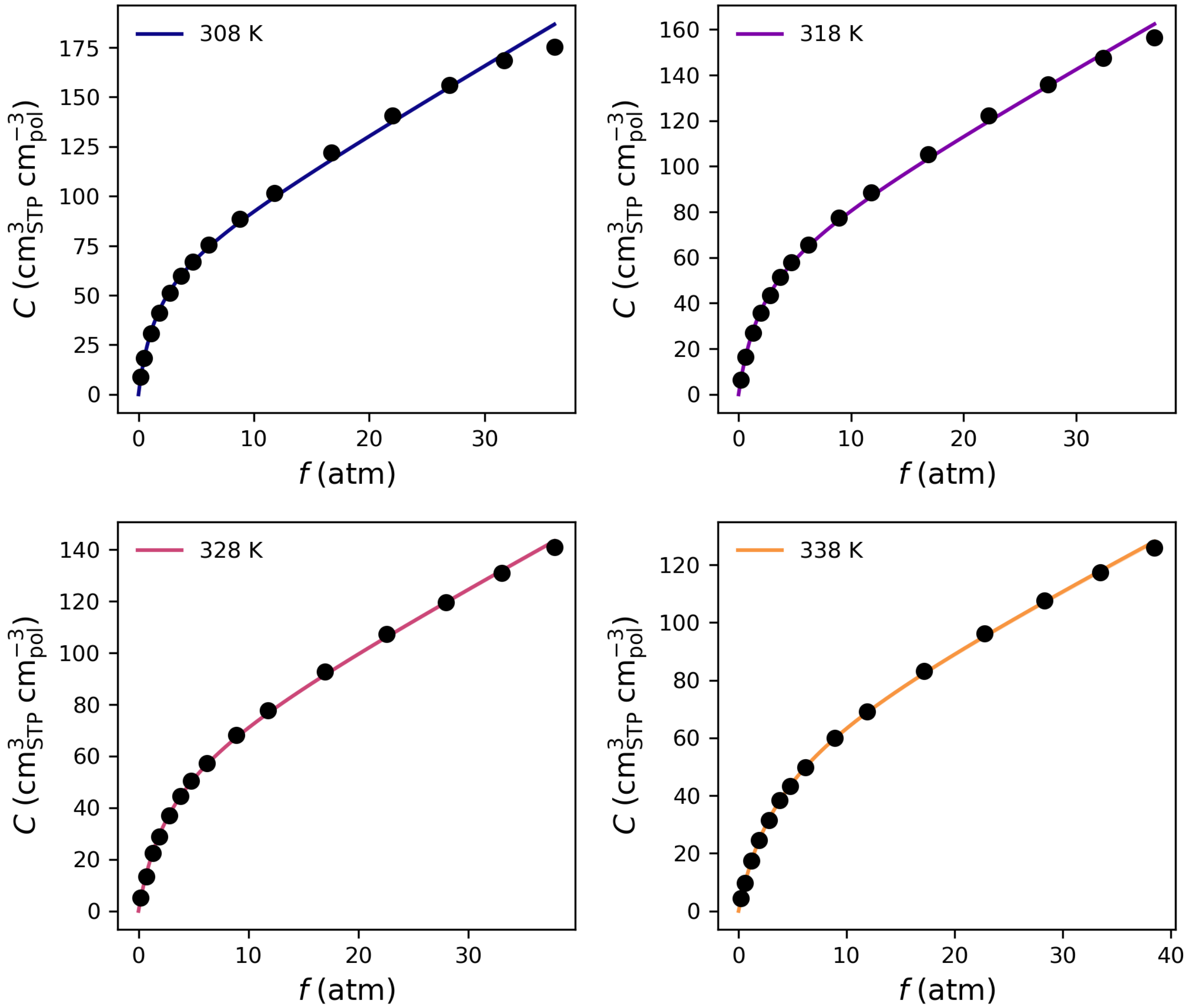
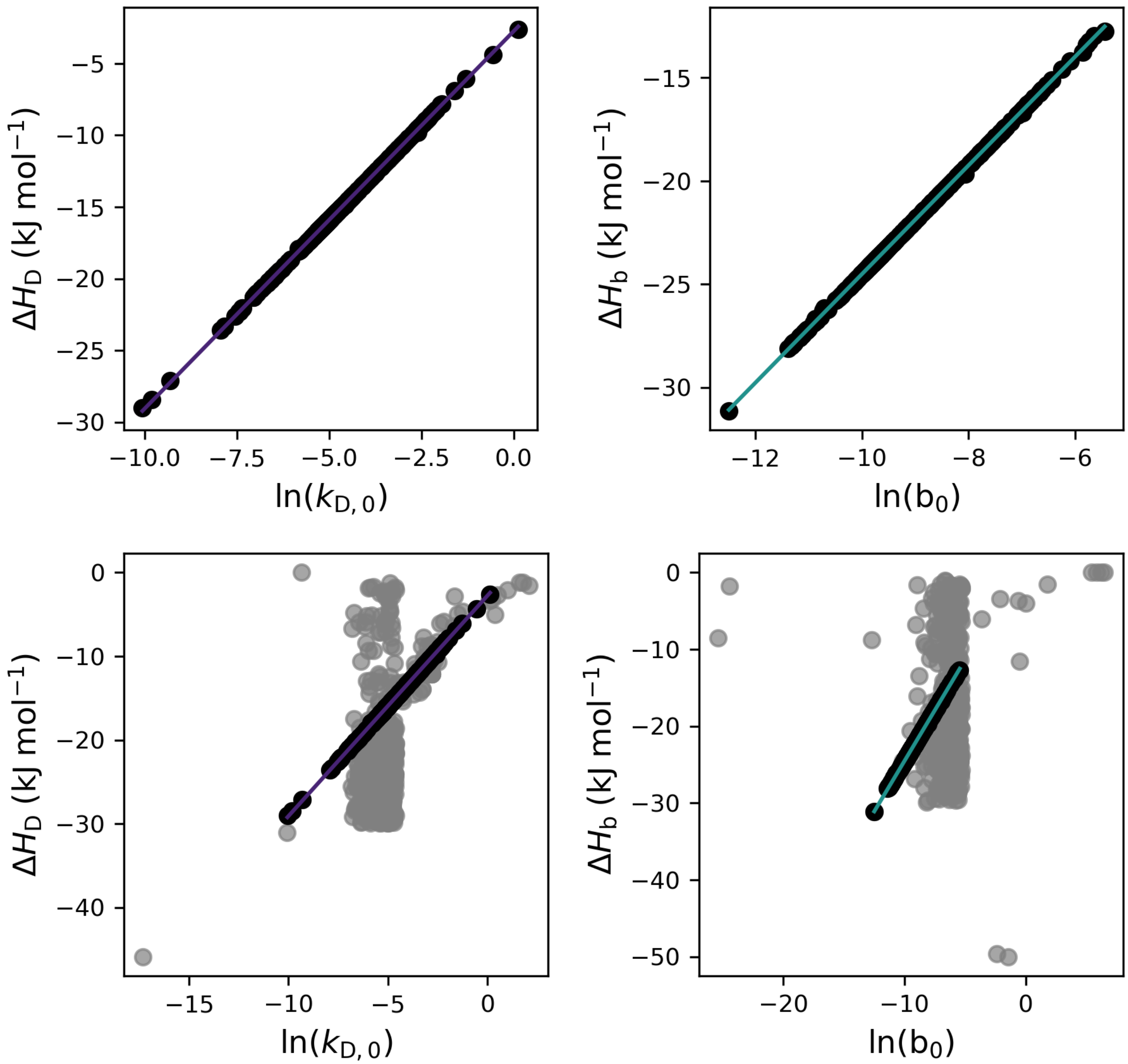
Sorption Isotherms

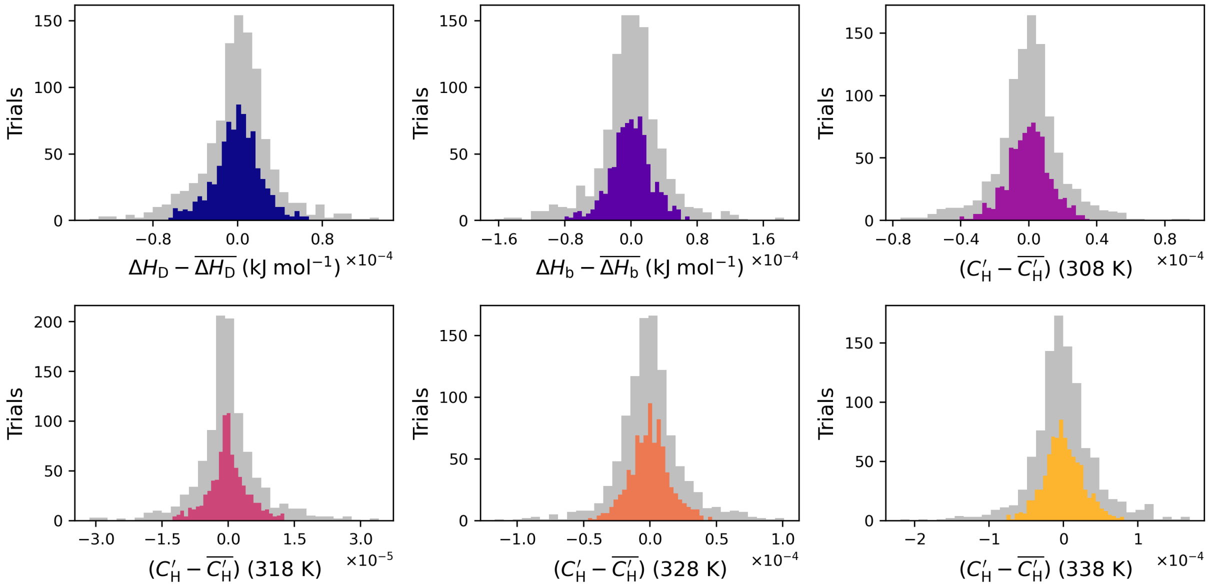


Linear Free Energy Relationships

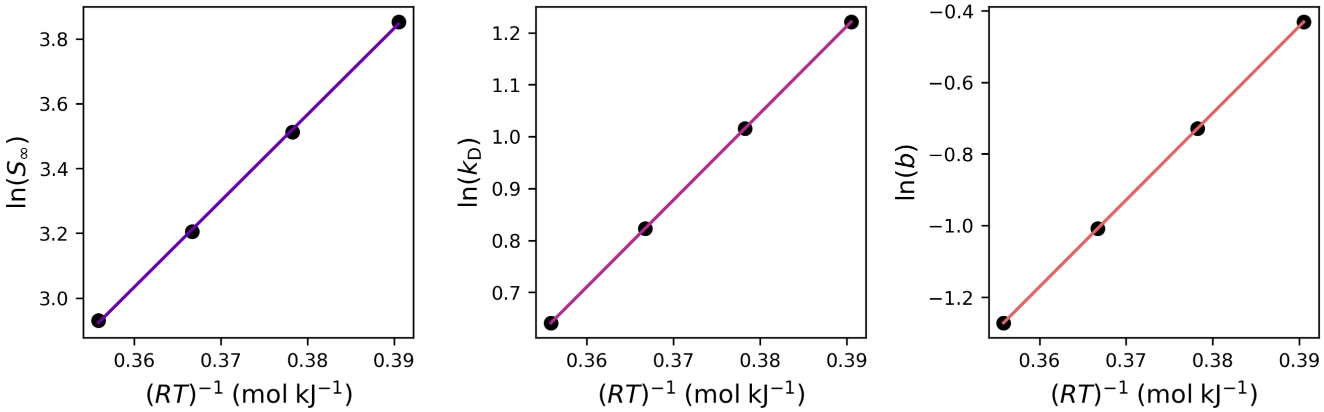
Heat of Henry Sorption (kJ/mol)	ΔH_D = -2.7444 + 2.6326 ln($k_{D,0}$)
Heat of Langmuir Sorption (kJ/mol)	ΔH_b = 1.8156 + 2.6325 ln(b_0)



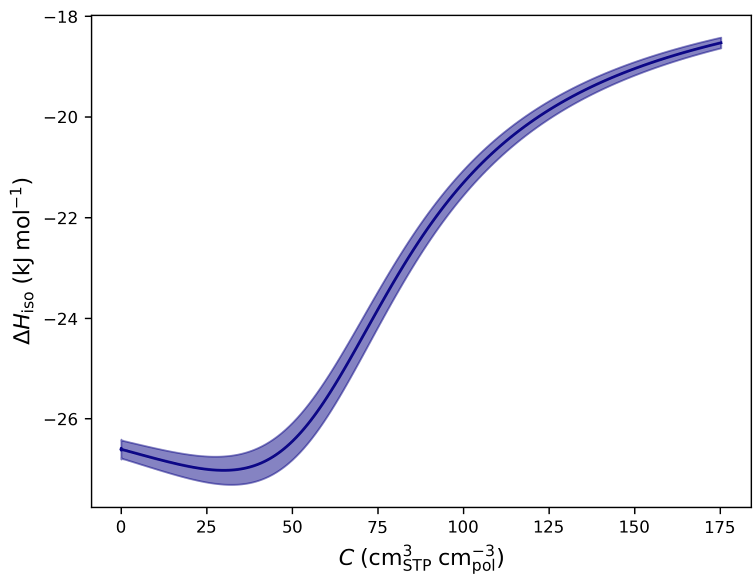
van't Hoff Relationships



Sorption Energetics



	Entropic Prefactor ($S_{i,0}$)	ΔH_i (kJ/mol)
i = Infinite Dilution	1.44e-03±1.92e-04	-26.6052 ± 0.3581
i = Henry	4.95e-03 ± 8.19e-17	-16.7178 ± 0.3609
i = Langmuir	5.00e-05 ± 1.25e-18	-24.2536 ± 0.1598



Analysis Settings

Parameter	Value
ΔH_D Initial Guess Range	[-1, -30]
ΔH_b Initial Guess Range	[-1, -30]
$k_{D,0}$ Initial Guess Range	[0.001, 0.01]
b_0 Initial Guess Range	[0.0001, 0.005]
ΔH_D Solver Bounds	[-50, 0]
ΔH_b Solver Bounds	[-50, 0]
$k_{D,0}$ Solver Bounds	[0, None]
b_0 Solver Bounds	[0, None]
C'_H guess #1	[0 100]
C'_H guess #2	[0 100]
C'_H guess #3	[0 100]
C'_H guess #4	[0 100]
C'_H bounds #1	[0 150]
C'_H bounds #2	[0 150]
C'_H bounds #3	[0 150]
C'_H bounds #4	[0 150]
Trials	1000
LFER Solver	SLSQP
maxiter (LFER solver)	1000
van't Hoff Solver	SLSQP
maxiter (van't Hoff solver)	1000
ftol (SLSQP)	1e-07
xtol (trust-constr)	1e-07
gtol (trust-constr)	1e-07
Seed	None

Questions or Comments?

Please reach out through the pyDMS GitHub page (*INSERT LINK*)

Citation

If you used pyDMS in any presented or published work please cite:
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